



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - Standard investigation

Drafted 04/10/2026

INVESTIGATION WORKFLOW SUITE

Contents

Executive summary	3
Event overview	5
Analysis	8
People	8
Environment	11
Equipment	13
Procedures	15
Organisation	17
Excluded hypotheses.....	19
Findings summary	21
Contributing factors	21
Other factors that increased risk	21
Other findings.....	21
Appendices.....	23
Appendix A: Summary of factor testing	23
Appendix B: Excluded hypotheses	24
Appendix C: Sources of evidence	25
Appendix D: Investigation activity	26
Distribution	26

Executive summary

At 02:47:12 local time on 12/08/2026, HT-407, a 220 t class haul truck descending empty, struck the right side of light vehicle LV-118 as it entered the main ramp of the Stage 4 open pit at Bindarra Gold Mine from the RL 312 dewatering spur. The impact, at 24 km/h, pushed LV-118 18.4 m down the ramp and rolled it onto its left side against the inner windrow.

The driver was a grade control technician on his third night shift at the mine. He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. He was extricated 27 minutes after the collision and flown to the metropolitan trauma centre at 07:40. The light vehicle was destroyed. The truck was damaged at the front left and returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes, and the ramp below RL 340 closed for 22 hours.

Why it happened

Both vehicles very likely arrived at the intersection with neither operator aware of the other. The spur met the ramp 34 m beyond a crest. This gave 41 m of sight line against a plan minimum of 120 m and a truck stopping distance of 96 m, which very likely contributed. The spur was opened under a works order classified as routine earthworks. No management of change was raised, so the spur never reached the traffic management plan, the in-cab map or the collision awareness system configuration, which very likely contributed. The plan required positive radio calls only at nine listed intersections and allowed the list to change only at the annual review, which very likely contributed. Neither operator made a call, which likely contributed.

LV-118 entered the ramp without stopping or establishing the truck's position, which very likely contributed. The intersection had no sign, line or give-way control, and a stop at the spur mouth would have added only a few metres of sight line. The driver's contractor induction was accepted as equivalent. The site supplement and pit driving assessment were never verified, although the site standard says they may not be substituted. This likely contributed.

The collision awareness system detected LV-118 at 21.4 m but raised no alert. The intersection lay inside a suppression geofence, which was extended in June 2026 without anyone outside the configuring group being told. This likely contributed. An alert would have come after braking began, so the finding rests on months of silence on that ramp having shaped the operator's expectations. The external guidance left the review trigger undefined and the interaction zones to the site, which likely contributed. That finding is held with less confidence than the others. It reflects the breadth of discretion in the guidance, not any breach by the mine or failure by the Inspectorate.

The truck operator braked and steered towards the wall side within 1.4 seconds of sighting. This reduced the impact speed from 37 km/h to 24 km/h and kept LV-118 away from the drop.

Other factors

Night shift extrication depended on one rostered paramedic and a second responder called from the village 11 km away. This delayed extrication by approximately 9 minutes, which did not contribute but very likely increased risk. The operator was on night 10 of 14, in the circadian low. This roster exposure likely increased risk but was not shown to have influenced the occurrence. Whether dust reduced visibility could not be determined. Wet surface and degraded brakes were ruled out.

Safety issues and actions

Five safety issues were identified: the works order exclusion from management of change, the closed listing route for positive calls, the unreported geofence change, the unverified pit driving assessment and the external guidance's undefined triggers. The night shift emergency response arrangement is a sixth.

At 05:10 on 12/08/2026 the Mining Manager directed a positive call before entering any Stage 4 intersection. The RL 312 spur was closed. The other ten unlisted Stage 4 roads were surveyed, and two that were below the minimum sight distance were closed pending a stop control. Warrigal technical services began reporting suppressed proximity events. The investigation recommended:

- a route by which intersections reach the plan between reviews, with mandatory referral of new roads
- removal of the works order exclusion, or carriage of its three referrals on the works order
- treatment of geofence changes as changes to a documented control, with operators told where alerts are suppressed

Event overview

At 02:47:12 local time on 12/08/2026 (18:47:12 UTC on 11/08/2026), a haul truck descending the main ramp of the Stage 4 open pit at Bindarra Gold Mine collided with a light vehicle entering from the RL 312 dewatering spur. The light vehicle's driver was seriously injured.

The truck, HT-407, was a 220 t class rigid haul truck operated by Warrigal Mining Services. Its operator had 7 years on the type and 14 months at Bindarra. He was on night 10 of a 14 night swing, 10 hours 47 minutes into a 12 hour 30 minute shift, and had slept 6 hours 30 minutes in the preceding day. The light vehicle, LV-118, was driven by a grade control technician employed by the survey subcontractor Merrivale Survey & Geotechnical. He had 6 years in grade control and was on his third night shift at Bindarra.

At RL 312 the main ramp runs approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, with a dry, well maintained surface. The spur joins from the east, on the inside of a left-hand bend, at an included angle of 71 degrees, 34 m beyond the high point of a crest. From the truck's cab (eye height 5.80 m) the spur mouth first comes into view at 41 m. From the spur mouth (eye height 1.45 m) a descending truck first comes into view at 38 m. A 1.6 m inner windrow separates the last 25 m of the spur from the ramp.

Accounts of visibility differ. The truck operator said some dust was hanging and he could not say whether it made any difference. The light vehicle driver's preliminary account was that the air was clear and he could see the ramp surface and the windrow.

The spur and the week before

The spur is a 240 m access road to dewatering bore DW-14, cut under works order WO-26-4471 as routine earthworks not requiring management of change. The earthworks were recorded as complete on 31/07/2026. On 02/08/2026 the bore was energised, light vehicles first used the spur, and the "ROAD CLOSED - CONSTRUCTION" sign at the spur mouth was removed. Nothing replaced it.

On the night of the occurrence the intersection had no warning sign on either ramp approach, no give-way or stop control, no delineation and no posted speed. The fleet management system in-cab map for the Stage 4 pit, last published on 24/07/2026, did not show the spur. The intersection was not one of the nine at which the traffic management plan requires a positive radio call.

At about 22:10 on 04/08/2026, a dewatering technician came out of the spur in front of a descending haul truck, reversed and let it pass. That night he lodged hazard report HAZ-2026-0917, which stated: "you cannot see up the ramp from that road. There is a rise in the ramp and a windrow along the side and you are out on the road before you can see anything." It also stated: "I do not think they know the road is there."

The Dewatering Superintendent closed the report on 06/08/2026, recording that the crew had been reminded at the pre-start on 05/08 "to take care entering the main ramp and to give way to haul trucks at all times." The risk rating was reduced from Medium to Low, and the report was not referred outside the dewatering department.

On 09/08/2026 the light vehicle driver was issued a site access card at the gatehouse against Merrivale's recognition agreement, after a 20 minute briefing on muster points, speed limits and personal protective equipment. His pre-arrival induction was a two hour online module completed on 24/07/2026. There is no evidence that he completed the Bindarra site-specific supplement or the light vehicle pit driving assessment. His first Bindarra night shift began that evening.

The night shift

At 02:22:41 dispatch called the Stage 4 crews back from crib. At 02:38:55, after its first post-crib load, HT-407 reported "407 dumped, coming back down." At 02:46:18 dispatch told the excavator operator that HT-407 was "about four minutes off you."

The light vehicle driver had been taking levels at the DW-14 collar. Nobody on the shift knew he was in the pit, as there was no permit, call-in or board by which they could have known. He said he left the collar at "about ten to three", later than the recorded collision. From the 240 m spur length and his position when the vehicles first came into sight, the investigation placed his departure at about 02:45 and his arrival at the spur mouth at 02:47.

His preliminary account is that he slowed at the mouth, looked left up the ramp, saw nothing and drove out at an estimated 25 km/h. On read-back he corrected "stopped" to "slowed", and said he did not believe he came to a complete stop. His radio was on channel 12, as the dewatering foreman had instructed. He had driven the spur six or seven times without meeting a truck.

The approach and collision

The truck descended at 41 km/h at RL 368 and 39 km/h at RL 348. At 02:47:06 it was at 37 km/h at RL 314, 41 m above the intersection. Coming over the crest, the operator saw the light vehicle's beacon whip at about 40 m, a distance he gave unprompted and which agrees with the survey to within a metre.

Neither driver made a radio call. At 02:47:07.4, 1.4 seconds after the last steady data point, service brake pressure rose from 0 to 620 kPa in 0.4 seconds, the retarder went to maximum and 14 degrees of right steering was applied. The operator said he braked and steered in one movement, right into the windrow, because to the left was a hundred metre drop with the light vehicle in front of him. The braked left front tyre left a continuous scuff beginning 38.5 m up-ramp of the spur mouth, deviating 1.4 m right over its first 20 m.

At 02:47:09.2 the truck's collision awareness system logged LV-118 at 21.4 m, closing at 13 km/h. No in-cab alert was raised, because the intersection lay inside the S4-LOAD suppression geofence.

At 02:47:12 the truck's front left struck the right side of LV-118 at 24 km/h, 6.2 m into the ramp from the spur mouth. Mean deceleration from brake application to impact was 1.85 m/s². LV-118 was pushed 18.4 m down-ramp, its marks changing from tyre scuff to body panel gouge at 11.2 m where it went over. It came to rest on its left side against the inner windrow, facing down-ramp. The truck stopped 19 m below the intersection at 02:47:14.3.

The light vehicle's roll-over protection structure was deformed inward at the right rear mounting but not breached. The driver's seatbelt was found fastened, with loading marks on the webbing.

The response

At 02:47:41 the truck operator called: "Dispatch, 407. I've hit a light vehicle on the ramp at three-twelve." At 02:48:01, 49 seconds after the collision, dispatch broadcast: "All Stage 4 units stop. All units stop." The operator stayed in his cab until cleared to get out at 02:48:44.

The emergency response team was paged at 02:48. At 02:49:30 the operator reported: "He's conscious. He's talking to me. He can't get out, the door's against the ground." The supervisor, travelling from the ROM pad, directed that the driver not be moved.

The first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The night shift is rostered with one paramedic, and a second responder is required before any extrication. The second responder came from the accommodation village, 11 km from the gatehouse, arriving at 03:07 with the hydraulic rescue tool. The driver was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision. The post-incident review found the extrication was delayed by approximately 9 minutes and that the delay did not affect the clinical outcome on this occasion.

The driver reached the surface medical facility at 03:41 with a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was flown to the metropolitan trauma centre at 07:40.

The light vehicle was destroyed. The truck, damaged at the front left, returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes and the ramp below RL 340 was closed for 22 hours. The truck operator and the shift supervisor both returned negative drug and alcohol tests. At 05:10 the Mining Manager directed that a positive call be made before entering any Stage 4 pit intersection.

Analysis

The collision at the RL 312 spur intersection came down to one condition. HT-407 and LV-118 arrived at the same intersection, and neither operator knew the other was there. This analysis treats that convergence as a single mechanism rather than a series of separate failures. Four conditions each removed a way in which one operator might have learned of the other:

- the geometry of the intersection
- the absence of any radio call
- the suppression of the collision awareness alert
- the manner of the light vehicle's entry onto the ramp.

The procedural and organisational conditions examined later in this section explain why each of those means was missing on the night.

HT-407 was descending empty at 37 km/h. Its in-cab road network showed no intersection at RL 312, and there was no warning sign on its approach. Nothing about the spur or the light vehicle was said on the pit channel. Its collision awareness system detected the light vehicle at 21.4 m and presented nothing in the cab.

LV-118 was leaving the spur at about 25 km/h. From the spur mouth its driver had 38 m of sight line up the ramp. There was no give-way control, he made no call, and he did not know the truck was there.

The first sighting available to either operator was at 41 m. At 37 km/h that is about four seconds of travel, which is what the survey of the intersection predicts for its geometry. None of the following gave any indication that either operator knew of the other before that moment:

- the pit channel recording
- the fleet management system download
- the collision awareness system log
- the interviews with the truck operator, the light vehicle driver and the night shift supervisor.

The night shift supervisor did not know a light vehicle was in the pit at all. There was no permit, call-in or board by which he could have known.

This condition was assessed as extremely likely rather than virtually certain to have existed. LV-118 carried no fleet management system and only a passive collision awareness tag, so its movements before the last two seconds depend on the driver's account alone. Once the two vehicles were converging with 41 m of sight line, nothing between that condition and the collision could have prevented it.

It was extremely likely that both vehicles arrived at the intersection with neither operator aware of the other, and extremely likely that this contributed to the collision.

People

The light vehicle's entry onto the ramp

The light vehicle driver's account is that he came out of the spur at about 25 km/h, looked left up the ramp, saw nothing and drove out. At interview he first said he had stopped. On read-back he corrected this to slowed, and said he did not believe he had come to a complete stop. That correction ran against his own interest. His account agrees with the scene evidence: the impact was 6.2 m into the ramp from the eastern edge of the spur mouth and 3.1 m out from the toe of the inner windrow.

BIN-TMP-004 clause 4.1 states: "Light vehicles shall give way to heavy vehicles at all times and in all circumstances." The driver described himself as applying that rule. Nothing at the spur mouth prompted him to do anything more. There was no give-way control, no stop line, no warning sign and no posted speed. Clause 6.4 of the plan applies only to intersections shown on the road layout drawing at Schedule 1, and the spur was not on it. The driver had not completed the Bindarra light vehicle pit driving assessment and had never seen the traffic management plan. The organisational reasons for that are examined under Organisation.

The survey limits what a different approach at the mouth could have achieved. From the mouth, at the driver's eye height of 1.45 m, a descending truck first comes into view at 38 m. The 1.6 m inner windrow hides the truck's lower body for the last 25 m of the spur's approach. A look left from a stationary position at the mouth would therefore have shown him about what a look at 25 km/h showed him. A full stop would have gained a few metres of sight line, not the 96 m a truck at the posted speed needs to stop.

The investigation records this as an action, not a failing. The driver was on his third night shift at the mine. Nothing at the intersection, and nothing in the induction he had been given, indicated that this intersection required more than the give-way he applied.

Even so, entering the ramp without establishing the truck's position was the step that placed the light vehicle across the truck's path. Without the light vehicle on the ramp there would have been no collision. The influence of this action was discounted from certainty because, as the survey shows, a full stop alone would not have given the driver a view of the truck in time.

It was virtually certain that the light vehicle entered the ramp from the spur without stopping and without establishing the truck's position, and very likely that this contributed to the collision.

The absence of a radio call

The pit channel was logged continuously. The investigator listened to the whole period from the return from crib to two minutes after the collision. Neither vehicle made any transmission about the RL 312 spur, the dewatering bore, or a light vehicle being anywhere in the Stage 4 pit. Apart from logging on at the start of shift, LV-118 made no transmission on the channel that shift.

No call was required. BIN-TMP-004 clause 4.3.2 states: "At intersections not listed at Appendix B, vehicles shall give way in accordance with clause 4.1. No call is required." The RL 312 spur intersection is not listed at Appendix B.

The truck operator, the light vehicle driver and the night shift supervisor each described the same practice, independently and without being led. Calls are made at the nine intersections on the crib room board and nowhere else. The supervisor said he had never corrected anyone for not calling elsewhere. He considered that requiring calls at every intersection would make the channel unusable.

The truck operator's training was consistent with that practice. His general induction, in 06/2025, included a 23 slide traffic management module. It named the nine Appendix B intersections and did not say what to do at an intersection that was not listed. He had had no traffic management refresher since.

This omission therefore has no owner. The plan expressly required no call at this intersection, the local practice matched the plan exactly, and neither operator departed from what the site expected of them.

A call would very probably have made a difference. A transmission from either vehicle would have been heard by the other vehicle, by dispatch and by the supervisor. Dispatch was already following the truck closely: at 02:46:18 the dispatcher told the excavator operator "407's about four minutes

off you." The influence was discounted from certainty because a call gives knowledge, not separation. Even with a call made, the two vehicles would still have had to resolve who went first at an intersection with 41 m of sight line.

It was virtually certain that neither operator made a positive call on the pit channel before entering the intersection, and likely that this contributed to the collision.

The truck operator's response

The fleet management system download shows that the operator's response reduced the severity of the outcome. Its last steady data point was at 02:47:06, with the truck at 37 km/h and 41 m above the intersection. From 02:47:07.4, three things happened in the same movement:

- service brake pressure rose from 0 to 620 kPa in 0.4 seconds
- the retarder was taken to maximum
- 14 degrees of right steering was applied.

That is a response time of 1.4 seconds. It is inside the 1.5 seconds assumed in the stopping distance calculation for the intersection.

The download gives a mean deceleration from brake application to collision of 1.85 m/s², against 1.9 m/s² measured for the unit eight days earlier. The impact speed was 24 km/h. With no braking it would have been 37 km/h.

The scene marks confirm the steering input. The braked scuff deviates 1.4 m to the right over its first 20 m, towards the wall side of the ramp. The operator explained his choice: to the left was a drop of about a hundred metres, with the light vehicle in front of him. The steering kept the collision against the inner windrow, and the light vehicle came to rest there rather than going over the edge.

After the collision the operator stopped 19 m below the intersection and reported the collision on the pit channel. He stayed in his cab, as dispatch instructed, until the ramp was clear. He then went to the light vehicle and stayed with the driver until the response team arrived.

The 41 m sight line meant the collision could not be avoided. Within that constraint, the speed and direction of the operator's response are the reason this occurrence ended in a survivable injury.

It was virtually certain that the truck operator braked and steered into the windrow within 1.4 seconds of sighting, and very likely that this reduced the severity of the outcome.

Fatigue

The truck operator was on night 10 of a 14 night swing, working 18:00 to 06:30 local time. The collision occurred at 02:47, 10 hours 47 minutes into the shift and inside the window of lowest circadian alertness. Every limit in the contractor's fatigue management standard was met:

- no more than 14 consecutive night shifts
- a minimum 10 hour break between shifts
- no shift in the preceding nine extended beyond 12 hours 40 minutes.

His own sleep diary, which he kept voluntarily and provided to the investigation, showed 6 hours 30 minutes of sleep in the day before the occurrence. It showed between 5 and 7 hours on each of the preceding nine days. In interview he described himself as tired in the way an operator is on a tenth night, but not falling asleep, and awake and driving properly.

Fatigue at this point in a roster would be expected to degrade detection and slow response. Detecting a hazard in an unlit pit is the kind of task fatigue degrades first. The evidence did not support that mechanism here:

- **Response time.** The download gives a 1.4 second response, inside the 1.5 seconds assumed in the stopping distance calculation.
- **Quality of the response.** The braking and steering that followed were correct, immediate, and in the only direction available.
- **Detection.** The operator saw the light vehicle at the distance the geometry allows.
- **Margin.** The investigation's analysis of the geometry indicated that, at 41 m of sight line and 37 km/h, even an immediate response would not have avoided the collision.

The download shows no decision or omission through which fatigue could have acted. The influence of fatigue on this occurrence was assessed as unlikely.

The exposure is not confined to one night. The Stage 4 night roster places every haul truck operator at the controls of a 220 t class truck in the early hours of a tenth consecutive night. There is no fixed lighting anywhere in the pit. At an unlisted intersection, the only defence against an unlit hazard is the operator's own eyes. The same roster is worked on three other pits at the site.

It was virtually certain that the operator was working night 10 of 14, in the eleventh hour of a 12 hour 30 minute shift, in the circadian low. It was unlikely that this influenced the occurrence, and likely that it increased risk.

Environment

The sight line at the intersection

The mine survey department surveyed the intersection, the crest and the eye positions of both vehicles on 13/08/2026. In the survey's words, "The spur meets the ramp 34 m beyond the high point of a crest in the ramp." An operator descending at an eye height of 5.80 m first has line of sight to the spur mouth at 41 m. A driver at the spur mouth, at an eye height of 1.45 m, first has line of sight to a descending truck at 38 m.

BIN-TMP-004 clause 6.2 states: "At every intersection, the available sight distance shall be not less than the stopping distance of the largest vehicle using the road at the posted speed, and in no case less than 120 m. Where this cannot be achieved, the intersection shall be controlled by a stop control, signalised control or traffic controller."

The survey calculated the stopping distance of a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade. Using the manufacturer's data and a 1.5 second response, it is 96 m. The survey concluded: "The available sight distance at this intersection is 41 m. The intersection does not meet clause 6.2 on either limb." The 41 m available was less than half the 96 m stopping distance and about a third of the plan's 120 m minimum. None of the controls the plan requires in that case was in place.

The scene examination found that, at the time of the occurrence, the intersection carried:

- "no intersection warning sign on either ramp approach;
- no give-way or stop control at the spur mouth;
- no delineation, reflective marker posts or windrow gap markers;
- no posted speed for the spur itself."

A "ROAD CLOSED - CONSTRUCTION" sign had stood at the spur mouth while the road was being cut. It was removed on 02/08/2026, the day the spur was first used, and nothing was put in its place. The intersection did not attract the clause 6.4 signage, give-way control and delineation because it was not on the road layout drawing to which clause 6.4 applies.

The same sight line had been reported seven days earlier. A dewatering technician wrote in HAZ-2026-0917: "The problem is you cannot see up the ramp from that road. There is a rise in the ramp

and a windrow along the side and you are out on the road before you can see anything." His account and the survey measurement describe the same geometry independently.

The 41 m figure has to be read against the 96 m stopping distance. A truck at the posted speed could not stop for anything at this intersection, however alert its operator and whatever the operator did. Attention, training and radio calls could only have helped by keeping the two vehicles from arriving at the intersection together. None of them could have made the intersection safe once both were committed to it.

The influence of the sight line was discounted from certainty for one reason. A short sight line produces a collision only once something is in the way, and it was the other conditions examined in this section that put the two vehicles there together.

It was virtually certain that the spur met the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m, and very likely that this contributed to the collision.

Light, surface and dust

The occurrence was in darkness. The light, surface and weather records state: "There is no fixed lighting anywhere in the Stage 4 pit; the only lighting is vehicle lighting and the light plants at the loading face, 1.1 km away and out of sight behind the crest." Moonset had been at 23:18 the previous evening. The ramp surface was dry. The question of the surface is addressed with the excluded hypotheses at the end of this section.

The investigation examined whether dust on the ramp reduced the truck operator's ability to see the light vehicle. That question could not be resolved.

Evidence that dust was present. The water cart log shows the main ramp between RL 280 and RL 400 was last watered at 01:10, 1 hour 37 minutes before the occurrence. The standing night shift instruction is a pass every 90 minutes, so a pass was 7 minutes overdue. The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre in the 02:40 to 02:50 interval, against a shift mean of 88. The truck operator described dust hanging on the ramp. He said the cart had been through earlier and that the surface dries out. His descent was behind his own lights into an unlit pit, which is the condition in which airborne dust is worst for a driver.

Evidence against an effect on visibility. The light vehicle driver was in the same place four seconds earlier. He was firm that the air was clear, that he could see the ramp surface and the windrow, and that the dust at Bindarra is nothing like that at the other sites where he works. The truck operator himself declined to say that the dust made any difference. The PM10 reading of 142 was well below the site trigger of 250. It measured airborne particulate at a monitor head 180 m away, not what could be seen from either cab. Nothing at Bindarra measures visibility.

The strongest point against is the sighting distance. The truck operator saw the light vehicle at the distance the survey predicts for the geometry in clear air. His account and the sight line agree without any reduction in visibility being needed to explain them.

Why the question could not be resolved. The deviation in particulate levels was real, and it occurred in the right place at the right time. But the two first-hand accounts did not agree, and the one available measurement showed a deviation, not an effect on what could be seen. Settling the question would need a measure of visibility at the intersection, from the cab positions, on the night. No such measure exists, and none can be reconstructed now.

The existence of a dust effect on the truck operator's view was assessed as about as likely as not. This matter remains unresolved. It is not a finding, and it is not carried into the findings of this report.

Equipment

The collision awareness system and its suppression geofence

The configuration found on HT-407 was described as follows: "System operating at Level 7 (operator advisory). Level 9 (automated intervention) is available on this fleet and is not enabled at Bindarra. Advisory alerts are raised on a vehicle tag entering the 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence."

The suppression geofence S4-LOAD was drawn on 14/03/2025 around the Stage 4 loading face. Its purpose was to stop nuisance alerts between the excavator, the trucks under it and the service vehicles working in the same area, where the alert rate had made the system unusable.

The geofence was extended northward on 09/06/2026 when the loading face advanced. The extended boundary lies 180 m up-ramp of the RL 312 spur intersection, so the intersection sat inside the suppressed area. The contractor's technical services group made the extension as a system configuration task. It was not recorded in the traffic management plan and was not reviewed against it.

The event log for the night shows the system detecting the light vehicle and suppressing the alert:

- "18:47:09.200 PROXIMITY HT-407 tag LV-118 range 21.4 m closing 13 km/h zone S4-LOAD alert SUPPRESSED (geofence)" [collision awareness system event log, HT-407, UTC]

That event is 02:47:09.2 local time. Outside the geofence, the same event would have raised an advisory tone and a screen indication in the cab.

The fleet-wide event log for the twelve months before the occurrence shows a pattern:

- HT-407 raised 412 operator advisory alerts. Every one was outside the S4-LOAD geofence and none was inside it.
- Across the Stage 4 haul fleet, 68 proximity events inside the geofence were logged and suppressed.
- Nine of those 68 involved a light vehicle tag on the main ramp below the geofence boundary. That is the same configuration of vehicles and location as this occurrence.

The suppressed events were stored in the system and visible to anyone who queried it. Nobody had queried it. The technical services group's standing report covers alert counts and system health and does not report suppressed events at all. No training record anywhere on site, for any person, covered the suppression geofences.

There were two readings of the suppression's influence on this collision.

Against an influence. The alert would have sounded at 02:47:09.2. That was 1.8 seconds after the operator had begun braking and 2.8 seconds before impact, so on this occasion it would have told him something he already knew. The system is advisory only, with no automated intervention. With 41 m of sight line, the outcome was fixed before either the alert or the sighting. On this reading, the suppression is a serious finding about the fleet but not a cause of this collision.

For an influence. The influence lies not in the alert that did not sound in the last three seconds, but in what the operator had come to expect. His experience of the system on that ramp was that it stayed silent, and he had come to read that silence as meaning the road was empty. He raised this in his own account before he was asked about it. The supervisor also raised it unprompted, and the alert statistics bear both of them out. That expectation was a condition the operator carried over the crest.

The investigation accepted the influence on the second reading and assessed it as likely. That is a lower level of confidence than for most of the other contributing factors. The influence is inferred

from the operator's account and the alert statistics, not shown by the recorded data, and both readings are carried here for that reason.

The existence of the geofence is not the problem. Nuisance alerts around an excavator would render the system unusable, which is why the geofence was drawn. The problem is that its boundary moved 180 m in June 2026 and nobody outside the group that moved it was told.

It was virtually certain that collision awareness alerts were suppressed inside a geofence that had been extended to cover the new intersection, and likely that this contributed to the collision.

The impact

The truck struck the light vehicle at 02:47:12.0 at 24 km/h, 4.6 seconds after brake application. The impact point was 6.2 m into the ramp from the eastern edge of the spur mouth and 3.1 m out from the toe of the inner windrow. The truck's accelerometer logged 1.9 g longitudinal.

The damage pattern matches the recorded data:

- **LV-118:** damage concentrated on the right side from the B pillar forward.
- **HT-407:** damage confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line.
- **Debris:** right-side glass and mirror fragments from 4.8 m to 14 m down-ramp of the impact position.

The geometry of the intersection made a side impact the likely form of any collision there: a light vehicle emerging from the spur crosses the path of a descending truck. When a 220 t class truck strikes the side of a light vehicle, the outcome depends on three things: the light vehicle's structure, the occupant's restraint, and the speed. The speed was reduced by the truck operator's response, but by the time of the sighting the impact could no longer be avoided.

It was virtually certain that the haul truck's front left struck the right side of the light vehicle at 24 km/h, and virtually certain that this contributed to the occurrence.

The roll-over and the protection of the occupant

A continuous gouge runs 18.4 m down-ramp from the impact position. At 11.2 m it changes from tyre scuff to body panel gouge, which is where the light vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp. The door on that side was against the ground, so the driver was extricated through the rear of the vehicle.

The driver was injured but conscious throughout. He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. The roll-over is how the collision produced the injuries that make this a serious occurrence.

Two defences worked:

- The roll-over protection structure was deformed inward at the right rear mounting but was not breached.
- The driver's seatbelt was found fastened, with loading marks on the webbing.

The injuries are consistent with a side impact and roll-over in which the structure held and the restraint was worn. Had either of those been absent, the outcome would very probably have been different.

The light vehicle was serviceable. The inspection sheet for the week of 09/08/2026 recorded all items serviceable. The 4.5 m amber beacon whip was sheared at its base in the roll-over, and its lamp worked when tested after recovery.

It was virtually certain that the light vehicle was pushed down the ramp and rolled onto its left side, and virtually certain that this contributed to the driver's injuries.

Procedures

Positive radio calls at listed intersections only

The traffic management plan, BIN-TMP-004, contains the mine's own defence against exactly this interaction. Clause 4.3.1 states: "A positive call shall be made on the relevant pit channel before entering any intersection listed at Appendix B, and entry shall not be made until the call is acknowledged or the channel is confirmed clear." Clause 4.3.2 requires no call at any other intersection. Appendix B lists nine intersections.

The note to Appendix B sets out how an intersection reaches the list: "This appendix is reviewed and reissued at the annual review of this plan (clause 9.1). An intersection is added to this appendix where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication." The road layout drawing at Schedule 1 is dated 12/11/2025 and is also reissued only at the annual review.

No provision anywhere in the plan allows an intersection to be added to Appendix B, or to Schedule 1, between annual reviews. No provision requires a new road to be referred to the plan's owner when it is cut. The RL 312 spur appeared on neither document. The Appendix B board in the Stage 4 crib room was the 12/11/2025 issue.

The control itself is sound. Its trigger is circular: an intersection reaches the list only through a drawing that is reissued only at the review, and nothing requires a new road to be brought to that review at all.

Clause 9.1 does allow for more than the annual cycle: "This plan shall be reviewed annually, and following any significant change to the mine's road network." In practice that second limb had never been used. The document register shows annual reviews completed on 14/11/2023, 10/11/2024 and 12/11/2025. No out-of-cycle review had been held in the plan's eleven year history.

Since the 12/11/2025 review, 14 new roads had been cut in the Stage 4 pit. Three were cut under a management of change record and were referred to the plan's owner. The other eleven were cut under a works order, and none of those was referred.

The mine's hazard reporting did not close the gap either. A search of reports raised at Bindarra in the 24 months before the occurrence returned four that described the same kind of interaction: HAZ-2026-0917 and three others, from 03/2025, 09/2025 and 02/2026. Each of the three earlier reports described a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B, at three different locations. All four were closed within three days by the department that received them, with an entry recording that the crew had been reminded to take care and that heavy vehicles have right of way. None was referred to the plan's owner. None resulted in a change to Appendix B, to Schedule 1, or to any physical control.

The closure entry for HAZ-2026-0917, made by the Dewatering Superintendent on 06/08/2026, concluded: "Trucks have right of way on the ramp per the traffic management plan. No further action required. Closed."

Under the plan as written, the next date on which the spur could have been added to Appendix B was 12/11/2026, three months after the collision.

The influence of this gap on the occurrence was assessed as very likely. Had a call been required at the RL 312 intersection, the conditions for it to work were present: the pit channel was open, both vehicles were on it, and dispatch was following the truck closely enough to place it about four

minutes from the loading face. The influence was discounted from certainty because a call that is made but not acknowledged still leaves a driver deciding whether to go.

This is a safety issue.

It was virtually certain that positive radio calls were required only at listed intersections, with the list revised only once a year, and very likely that this contributed to the collision.

The external traffic management guidance

Mines Safety Guideline MSG-14 was issued by the State Mines Inspectorate under regulation 6.22 of the Mining Operations Safety Regulations 2018. Four of its clauses bear on this occurrence:

- **Clause 3.2** requires the traffic management plan to be "reviewed at intervals not exceeding twelve months, and following any significant change to the mine's road network, vehicle fleet or operating pattern."
- **Clause 3.5** states that "The plan is a site document. It is not approved by, and need not be submitted to, the Inspectorate, except on request following an occurrence or an inspection."
- **Clause 5.1** requires the operator to "nominate the intersections, crossings and other interaction zones at the mine at which positive communication between vehicles is required before entry."
- **Clause 5.4** requires a light vehicle to be "detectable by the heavy vehicle operator." It states that detectability "may be provided by line of sight, elevated marking, radio communication, a proximity detection system, or a combination of these."

The guideline does not define "significant change" and gives no example of one. It sets no minimum and no basis for nominating interaction zones. It does not say which, or how many, of the five means of detectability must be provided.

Bindarra's plan met the guideline on its face on the day of the occurrence. It had been reviewed within twelve months. It nominated nine interaction zones. It provided for detectability by line of sight and by elevated marking. The mine was not in breach of the guideline.

This finding is not a criticism of the Inspectorate. It concerns the shape of the discretion the guideline leaves to each site.

There were two readings of the guideline's influence.

For an influence. Three separate terms in the one guideline are left to the site to define. At Bindarra all three were defined in the same document, reviewed once a year, with no external scrutiny of any of them.

Against an influence. The guideline did not prevent anyone from acting. The site could have held an out-of-cycle review under its own clause 9.1 at any time, and MSG-14 does not stop a mine from listing every intersection it has. On this reading, Bindarra did not fail to act because the guideline was vague. It failed to act because nothing inside the mine ever put the question in front of the plan's owner, and the external guideline is background.

The investigation accepted the influence on the balance of these arguments. A defined trigger for review and a stated minimum for interaction zones would have given the eleven unreferred roads somewhere to go. Each of the site's own controls examined in this occurrence failed for want of exactly that. The influence was assessed as likely, which is a lower level of confidence than for most of the other contributing factors, and the question is not settled.

It was virtually certain that the traffic management guidance left the review trigger undefined and the nomination of interaction zones to the site, and likely that this contributed to the occurrence.

Organisation

The spur was built under a works order without management of change

The Corella Resources management of change procedure, CRL-PRO-022, sets out when a management of change record is required:

- **Clause 5.2:** "A management of change record shall be raised before any of the following: a change to the mine's road network; the introduction of a new vehicle type; a change to a pit design; a change to a documented control."
- **Clause 5.3** provides an exclusion: "Clause 5.2 does not apply to routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced. Such work is managed under the works order."
- **Clause 5.4** requires the change owner to complete an impact assessment. That assessment "includes referral to the traffic management plan owner, the survey department for the fleet management system road network, and the contractor's technical services group for the collision awareness system configuration."

Those three referrals exist only inside the management of change record. The works order form at Schedule 4 has no equivalent fields and no referral of any kind.

Works order WO-26-4471 was raised by the Mining Engineer on 28/07/2026 and approved by the Mining Superintendent the next day. Its scope was to "Cut and form 240 m of access road to dewatering bore DW-14 at RL 312." It classified the work as "Routine earthworks within the approved Stage 4 design. No new plant or vehicle type. Management of change not required, CRL-PRO-022 clause 5.3." Its handover field reads "N/A."

The works order contains none of the following:

- scope for signage
- a survey pickup of the new intersection
- a referral to the traffic management plan
- a request to publish a revised fleet management system road network
- a review of the collision awareness system geofences.

Because no management of change record was raised, none of the three referrals was made, and the three systems they would have reached were each left unchanged:

- **The traffic management plan.** The spur did not reach Schedule 1 or Appendix B, did not attract the clause 6.4 signage and give-way control, and was never assessed against the sight distance requirement in clause 6.2. It entered service with 41 m of sight line and none of the stop, signalised or traffic controller controls the plan requires in that case.
- **The in-cab map.** The fleet management system in-cab map for the Stage 4 pit, last published on 24/07/2026, was still current on the night. It showed the main ramp with no intersection at RL 312.
- **The collision awareness system.** The geofence configuration was not reviewed against the new intersection.

The pattern goes beyond this spur. Of the 14 new Stage 4 roads cut since the last plan review, the three built under a management of change record were referred to the plan's owner. None of the eleven built under a works order was.

The influence was assessed as very likely. A management of change record was the only route by which the new intersection could have reached the plan, the in-cab map or the geofence configuration, and without it all three stayed as they were. The influence was discounted from

certainty because a referral is not itself a control: the plan's owner might have considered the spur and still left it off the list.

This is a safety issue.

It was virtually certain that the access spur was cut and opened to traffic under a works order with no management of change raised, and very likely that this contributed to the occurrence.

Contractor induction and the pit driving assessment

The Corella Resources contractor management standard, CRL-STD-009, deals with contractor inductions in three clauses:

- **Clause 7.3** allows a contractor's own general induction to be "accepted as equivalent to the Bindarra general induction where the contractor holds a current recognition agreement with Corella Resources."
- **Clause 7.4** limits that equivalence: "Equivalence under 7.3 does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted."
- **Clause 7.5** places the check on the engaging department: "The engaging department is responsible for verifying that 7.4 has been completed before a site access card is issued."

Merrivale Survey & Geotechnical has held a recognition agreement since 2023. In practice, the gatehouse issues site access cards against the recognition agreement alone. It holds no record of the site supplement or the pit driving assessment and is not asked for one. Nothing in the access card system distinguishes a person who may drive in an operating pit from a person who may not.

The light vehicle driver's training file shows a two hour Merrivale online induction completed on 24/07/2026 and a 20 minute gatehouse briefing before his card was issued on 09/08/2026. It shows no record of the site supplement and no record of the pit driving assessment. He had never seen BIN-TMP-004 and did not know it existed.

He was driving a light vehicle on an active haul road on his third night at the mine. He had not completed the assessment that the site's own standard says may not be substituted, and no part of the system that admitted him was checking for it.

The standard itself is sound and says the right thing. What failed is that the only point at which verification could happen, the issue of the access card, has no field for it. This is not a failure by the driver.

The existence of this factor was assessed as extremely likely rather than virtually certain. The standard, the training files and the card register are documentary. However, Merrivale's own records of the site supplement were not examined, and the point was not put back to the driver.

The pit driving assessment is where a light vehicle driver is taught how the pit's traffic rules actually work: give-way, positive calls, and what to do at a blind intersection. Its influence on this occurrence was assessed as likely and discounted from certainty for two reasons. First, the main rule it would have taught, that light vehicles give way at all times, is one the driver was already applying. Second, the sight line at this intersection defeated careful observation on its own.

It was extremely likely that the contractor induction was accepted as equivalent with the site supplement and pit driving assessment never verified, and likely that this contributed to the occurrence.

Emergency response capability on night shift

The Bindarra night shift is rostered with one qualified mines rescue paramedic, based at the surface medical facility. A second responder must be present before any extrication begins, and is called in from the accommodation village 11 km from the gatehouse.

On the night, the response ran as follows:

- 02:48: dispatch paged the emergency response team.
- 02:51: the first response vehicle entered the pit.
- 02:54: the first response vehicle reached the scene.
- 03:07: the second responder arrived with the hydraulic rescue tool.
- 03:14: the driver was extricated, 27 minutes after the collision.

The post-incident review found that the extrication was delayed by approximately 9 minutes while the team waited for the second responder and the tool. It recorded that the delay did not affect the clinical outcome on this occasion, because the injuries were stable and the patient was conscious throughout.

This arrangement had no part in the occurrence. The collision, the injuries and their severity were complete before the emergency response team was paged. There was no path by which the response time could have affected the occurrence or its outcome, and the post-incident review reached the same conclusion on clinical grounds.

The arrangement is nonetheless a known, structural gap in a last line of defence, present on every night shift, and the site's own review has measured it at about nine minutes. On this occasion the driver was conscious, stable, restrained and protected by a roll-over structure that held, and the delay cost him nothing. A different casualty could have been affected:

- a fuel fire
- an unconscious occupant
- an airway that needed the vehicle opened in the first ten minutes.

In any of those cases, the same nine minutes could produce a different outcome. The site's extrication capability on night shift is a single person plus a drive from the village, and no part of the arrangement scales with the severity of what has happened.

This is a safety issue.

It was virtually certain that one emergency response paramedic was rostered on night shift with the second responder called from the village. It was unlikely that this influenced the occurrence, and very likely that it increased risk.

Excluded hypotheses

Two hypotheses were tested and excluded.

A wet ramp surface. This was assessed as very unlikely. The met station recorded no precipitation on the night, and the last rainfall at the site was 26 days before. The water cart's last pass on the section was at 01:10, 1 hour 37 minutes before the occurrence. The investigator found the ramp dry throughout at 06:20, with no damp patches, ponding or greasy film at or above the intersection. The braked tyre scuff ran continuously for 62 m with no sign of a locked or sliding wheel. The deceleration achieved, 1.85 m/s², was consistent with the unit's tested 1.9 m/s² on a dry 9 per cent grade.

Degraded service brakes. This was also assessed as very unlikely. Brake performance was consistent across three measurements:

- the monthly brake test on 03/08/2026, eight days before, gave 1.9 m/s²

- the occurrence download gave 1.85 m/s²
- the repeat test on 14/08/2026, before the unit returned to service, gave 1.92 m/s².

Service brake pressure rose to 620 kPa in 0.4 seconds, and the retarder was at maximum throughout. No brake, retarder or park brake defect had been raised against the unit in the twelve months before. The pre-start check sheet for 11/08/2026 was complete and recorded no defects.

Findings summary

A finding stated without a likelihood term was established as fact.

Contributing factors

- It is virtually certain that collision awareness alerts were suppressed inside a geofence extended to cover the new intersection, and it is likely that this suppression contributed to the occurrence. (Safety Issue)
- Traffic management guidance left the review trigger undefined and the interaction zones to the site, which was virtually certain, and the breadth of that discretion likely contributed to the occurrence. (Safety Issue)
- It is virtually certain that the access spur was cut and opened to traffic under a works order with no management of change raised, and it is very likely that this contributed to the occurrence. (Safety Issue)
- The spur met the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m, a condition that was virtually certain and very likely contributed to the collision.
- Under the traffic management plan, positive radio calls were required only at listed intersections and the list was rewritten once a year, an arrangement that was virtually certain and very likely contributed to the occurrence. (Safety Issue)
- It is virtually certain that neither operator made a positive call on the pit channel before entering the intersection, and it is likely that the absence of a call contributed to the occurrence.
- It is very likely that the contractor induction was accepted as equivalent with the site supplement and pit driving assessment never verified, and it is likely that this contributed to the occurrence. (Safety Issue)
- With virtual certainty, the light vehicle entered the ramp from the spur without stopping and without establishing the truck's position, and that action very likely contributed to the collision.
- Both vehicles very likely arrived at the intersection with neither operator aware of the other, and this convergence very likely contributed to the collision.
- That the haul truck's front left struck the right side of the light vehicle at 24 km/h was virtually certain, and it is virtually certain that the impact contributed to the driver's injuries.
- The investigation found it virtually certain that the light vehicle was pushed 18 m down the ramp and rolled onto its left side, and virtually certain that the roll-over contributed to the injuries the driver sustained.

Other factors that increased risk

- It is virtually certain that one emergency response paramedic was rostered on night shift, with the second responder called from the village, and it is very likely that this arrangement increased risk. (Safety Issue)
- The truck operator's exposure on night 10 of 14, in the eleventh hour of a 12.5 hour night shift and in the circadian low, was virtually certain, and it is likely that this roster exposure increased risk.

Other findings

- The investigation examined and ruled out the hypothesis that the ramp surface was wet and the truck could not stop in the distance available.
- Also examined and ruled out was the hypothesis that the haul truck's service brakes were degraded and could not achieve the required stopping performance.
- The truck operator braked and steered into the windrow within 1.4 seconds of sighting, and this response reduced risk.

Appendices

Appendix A: Summary of factor testing

Each factor examined by the investigation was tested in turn. The first test asked whether the factor existed. The second asked whether it influenced the occurrence. The third, applied where a factor did not influence the occurrence, asked whether it was nonetheless important to safety. The table gives the result and the confidence for each test, and the category that followed from them.

Factor	Lane	Existence	Influence	Importance	Category
Traffic management guidance left the review trigger undefined and the interaction zones to the site	External Influences	Pass, virtually certain	Pass, likely	Not tested	Contributing factor (Safety Issue)
Access spur cut and opened to traffic under a works order, with no management of change raised	Organisational Influences	Pass, virtually certain	Pass, very likely	Not tested	Contributing factor (Safety Issue)
Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified	Organisational Influences	Pass, extremely likely	Pass, likely	Not tested	Contributing factor (Safety Issue)
Positive radio calls required only at listed intersections, and the list rewritten once a year	Risk Controls	Pass, virtually certain	Pass, very likely	Not tested	Contributing factor (Safety Issue)
Collision awareness alerts suppressed inside a geofence extended to cover the new intersection	Risk Controls	Pass, virtually certain	Pass, likely	Not tested	Contributing factor (Safety Issue)
The spur met the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m	Local Conditions	Pass, virtually certain	Pass, very likely	Not tested	Contributing factor
Neither operator made a positive call on the pit channel before entering the intersection	Individual Actions	Pass, virtually certain	Pass, likely	Not tested	Contributing factor
The light vehicle entered the ramp from the spur without stopping and without establishing the truck's position	Individual Actions	Pass, virtually certain	Pass, very likely	Not tested	Contributing factor
Both vehicles arrived at the intersection with neither operator aware of the other	Occurrence Events	Pass, extremely likely	Pass, extremely likely	Not tested	Contributing factor

The haul truck's front left struck the right side of the light vehicle at 24 km/h	Occurrence Events	Pass, virtually certain	Pass, virtually certain	Not tested	Contributing factor
The light vehicle was pushed 18 m down the ramp and rolled onto its left side	Occurrence Events	Pass, virtually certain	Pass, virtually certain	Not tested	Contributing factor
One emergency response paramedic rostered on night shift, with the second responder called from the village	Risk Controls	Pass, virtually certain	Fail, unlikely	Pass, very likely	Other factor that increased risk (Safety Issue)
Night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low	Local Conditions	Pass, virtually certain	Fail, unlikely	Pass, likely	Other factor that increased risk
The truck operator braked and steered into the windrow within 1.4 seconds of sighting	Individual Actions	Pass, virtually certain	Pass, very likely (a decrease in risk)	Not tested	Positive factor
The ramp surface was wet and the truck could not stop in the distance available	Local Conditions	Fail, very unlikely	Not tested	Not tested	Excluded hypothesis
The haul truck's service brakes were degraded and could not achieve the required stopping performance	Occurrence Events	Fail, very unlikely	Not tested	Not tested	Excluded hypothesis
Dust on the ramp reduced the truck operator's ability to see the light vehicle	Local Conditions	Inconclusive, about as likely as not	Not tested	Not tested	Unresolved

The dust question is listed as unresolved because the evidence could not settle it either way. It is not a finding. Several things pointed towards dust. A water cart pass on that section was 7 minutes overdue against the standing 90 minute interval. The fixed monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre against a shift mean of 88. The truck operator also described dust hanging on the ramp. Several things pointed the other way. The light vehicle driver, in the same place about four seconds earlier, was firm that the air was clear. The truck operator declined to say that the dust made a difference. The reading was well below the site trigger of 250 and was taken at a monitor head 180 m away, not from either cab. Nothing at Bindarra measures visibility. The operator sighted the light vehicle at the distance the survey predicts for clear air, so the sighting is explained without dust. Settling the question would need a measure of what could actually be seen from the cab, and no such measure existed.

Appendix B: Excluded hypotheses

The ramp surface was wet and the truck could not stop in the distance available

This hypothesis was tested because the light vehicle was pushed 18 m down a 9 per cent grade, which raises a question about the surface. It was ruled out. The met station recorded no precipitation on the night, and the last rainfall at the site was 26 days before the occurrence. The water cart log puts the last pass on that section at 01:10, 1 hour 37 minutes before the collision.

The standing night shift instruction is a pass every 90 minutes, so the surface had been drying for longer than that interval and had not been freshly watered. The investigator examined the ramp at 06:20 local and found it dry throughout, with no damp patches, ponding or greasy film at or above the intersection. The braked tyre scuff ran continuously for 62 m with no sign of a locked or sliding wheel. The download gave a mean deceleration of 1.85 m/s², against the unit's own tested figure of 1.9 m/s² on a dry 9 per cent grade. No source gave any evidence of a wet or slippery surface.

The haul truck's service brakes were degraded and could not achieve the required stopping performance

This hypothesis was tested because a haul truck that does not stop in the distance available raises a question about its brakes. It was ruled out. The monthly brake performance test on 03/08/2026 gave a mean deceleration of 1.9 m/s² in the empty condition on a 9 per cent grade. The test was repeated on 14/08/2026, before the unit returned to service, and gave 1.92. The occurrence download gave 1.85 over the 4.6 seconds of braking. That figure sits between the two tests and within normal scatter. Service brake pressure rose from 0 to 620 kPa in 0.4 seconds, and the retarder was at maximum throughout. No brake, retarder or park brake defect had been raised against the unit in the twelve months before the occurrence. The pre-start check sheet for 11/08/2026 was complete and recorded no defects. The braking system performed to specification before, during and after the occurrence.

Appendix C: Sources of evidence

Interviews

The investigation interviewed three people. All three interviews had been completed by 21/08/2026.

- The haul truck operator. This interview was recorded and transcribed.
- The night shift supervisor. This interview was written up and read back to the interviewee.
- The light vehicle driver. This interview was written up and read back to the interviewee.

[NOT ESTABLISHED: the date of each individual interview]

Documents, data and other material

- Occurrence notification CRL-OCC-2026-0332, raised 12/08/2026 at 03:26 local
- Scene examination INV-2026-0332-SCENE, RL 312 intersection, examined 12/08/2026 before recovery, including signage and delineation
- Mine survey pickup SUR-26-1188 of the RL 312 intersection and sight distances, 13/08/2026
- Warrigal Mining Services technical services report WMS-TS-2026-114, issued 18/08/2026, including the fleet management system download for HT-407
- Collision awareness system configuration and event log for HT-407, extracted 12/08/2026
- Warrigal technical services fleet collision awareness system event log extract and alert statistics, 12 months to 11/08/2026
- Pit channel radio extract RAD-2026-0332, 02:20 to 02:52 local, transcribed 17/08/2026 and verified by a second investigator
- Bindarra Traffic Management Plan BIN-TMP-004 revision 7, issued 12/11/2025, clauses 4, 6 and 9 and Appendix B
- Document register for BIN-TMP-004 and Corella reporting system search, 24 months to 11/08/2026

- Hazard report HAZ-2026-0917, submitted 04/08/2026 and closed 06/08/2026, with its closure entry
- Management of Change procedure CRL-PRO-022 revision 3, clauses 5.2 to 5.4 and Schedule 4
- Works order WO-26-4471, RL 312 dewatering access, raised 28/07/2026 and completed 31/07/2026
- Contractor Management Standard CRL-STD-009 revision 5, clauses 7.3 to 7.5
- Warrigal training file, Merrivale training file and the Bindarra gatehouse access card register
- Warrigal roster and site access records, and the haul truck operator's sleep record, provided voluntarily
- Mines Safety Guideline MSG-14 revision 4, clauses 3.2, 3.5, 5.1 and 5.4, made under Mining Operations Safety Regulations 2018 r 6.22
- Met station, water cart log and fixed dust monitor records, night of 11 to 12/08/2026
- Warrigal asset register and Corella light vehicle register, including equipment maintenance, inspection and brake test records, obtained 13/08/2026
- Bindarra emergency response log and post-incident review, obtained 26/08/2026

Appendix D: Investigation activity

The times below are as logged, in UTC.

- 11/08/2026 at 20:15 UTC: The investigator was notified by the Night Shift Superintendent. The occurrence was assessed as a serious injury, reportable and warranting investigation. The investigation was opened and its scope written down. Statutory notification was made to the State Mines Inspectorate.
- 12/08/2026 at 04:00 UTC: The scene was attended before recovery. Both vehicles were photographed and measured in position, and the marks were recorded. The contractor took the fleet management and collision awareness downloads and handed them over sealed. Both operators were tested for drugs and alcohol, and the results were negative. A site-wide direction was issued overnight requiring a positive call at every Stage 4 intersection, listed or not.
- 13/08/2026 at 07:30 UTC: The survey pickup of the intersection was requested and received. It gave 41 m of available sight distance, against a plan minimum of 120 m and a stopping distance of 96 m. Works order WO-26-4471 was located, with management of change recorded as not required and the handover field reading N/A. Hazard report HAZ-2026-0917 was found in the reporting system and its closure entry was read.
- 21/08/2026 at 06:40 UTC: Interviews with the haul truck operator, the night shift supervisor and the light vehicle driver were completed. The collision awareness statistics were requested following the accounts given by the operator and the supervisor.
- 28/08/2026: A preliminary update was issued.
- 01/09/2026 at 02:00 UTC: The listing route for Appendix B of the traffic management plan was confirmed as closed between reviews. An intersection reaches the list only through a drawing reissued once a year, and nothing requires a new road to be referred at all. The matter was raised with the Mining Manager.

Distribution

Title	Name
-------	------

General Manager, Bindarra Gold Mine	T. Brandner
Registered Manager (statutory)	S. Kirkhope
Mining Manager and owner of BIN-TMP-004	H. Quaile
Manager Health and Safety, Corella Resources	L. Farquhar
Project Manager, Warrigal Mining Services	D. Mokoena
Superintendent Dewatering, Corella Resources	P. Nyaguse